

1 Question (25 points)

Design a TM that recognizes the language $L = \{a^n b^m c^m d^n | m, n \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

2 Question (25 points)

Design a TM that recognizes the language $L = \{a^n b^n c^n | n \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

3 Question (25 points)

Describe a multitape Turing machine with $k + 1$ tapes that recognizes the language $L = \{x_1^n x_2^n \dots x_k^n \mid n \geq 1\}$, $\Sigma = \{x_1, x_2, \dots, x_k\}$. You need to use all tapes in your solution. Do not design the TM. Describe the steps it needs to take.

4 Question (25 points)

Consider the language

$$L = \{ \langle D, k \rangle \mid D \text{ is DFA and it rejects all strings } w \text{ if } |w| < k \}, \text{ where } k \in \mathbb{N}$$

Prove that L is decidable by giving an TM algorithm.